Financial Resources, Transfer Spending and Sporting Performance at Arsenal Football Club, 2015/16 to 2024/25

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Type: Independent high-school research project
Period: 10 completed Premier League seasons (2015/16 to 2024/25)
Main sources: Companies House accounts of Arsenal Holdings, a census of the club's transfers, Premier League records and Understat

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ABSTRACT

I built a dataset of Arsenal's finances, transfers and results for ten seasons, 2015/16 to 2024/25, with 10 rows and 108 variables. The financial data come from the audited accounts of Arsenal Holdings filed at Companies House, the transfer data from a census of 128 transactions, and the match data from Premier League records and Understat. I tested six pre-specified sub-questions (SQ1 to SQ6) with simple correlations (Pearson, Spearman and Kendall), a leave-one-season-out check and a check that drops the COVID seasons.

Over the decade, football turnover rose by 97.09% (from £350.60m to £691.00m), staff costs rose by 77.58% (from £195.40m to £347.00m) and net debt rose from £6.50m to £283.00m. In the same season, transfer spending moved with turnover (r = 0.3659) and more closely with net debt (r = 0.5906). Across the nine pairs of consecutive seasons, spending in one season was more closely linked to the next season's expected goal difference (xGD, r = 0.5982) and expected points (xPTS, r = 0.6189) than to the next season's actual points per game (PPG, r = 0.3911), and much more closely than to expected points in the same season (r = 0.1535). Staff costs had almost no link to next-season PPG (r = 0.0516, Spearman = -0.0333), and the sign changed when single seasons were removed (range -0.0936 to 0.1648).

With only 10 seasons (9 lagged pairs), many tests, estimated transfer fees (77.8% of consideration comes from secondary reports) and a staff-cost figure that covers all employees, these results are patterns to test further. They do not show that spending causes results.

Keywords: sports economics, Arsenal FC, football finance, transfer spending, staff costs, expected goals

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1. INTRODUCTION AND RESEARCH DESIGN

European football has grown fast commercially, with rising broadcast income and rising spending on wages and transfer fees (Dobson & Goddard, 2011; Szymanski, 2015). Studies of many clubs find that wage spending is strongly linked to league position, and Hall et al. (2002) found evidence that in English football payroll spending tends to come before league performance (see also Szymanski & Smith, 1997).

Multi-club studies mostly compare rich and poor clubs. I wanted to see what happens inside one club when its money and results both change. Between 2015/16 and 2024/25 Arsenal went through several changes:

1. Three managerial eras: the last seasons of Arsène Wenger (2015/16 to 2017/18), Unai Emery with Freddie Ljungberg as interim manager (2018/19 to 2019/20), and Mikel Arteta (2019/20 to 2024/25).
2. A change in ownership and financing. Kroenke Sports & Entertainment (KSE) took full ownership in 2018, and the stadium bonds were later repaid (in 2020) and replaced by a loan from KSE.
3. Six seasons without Champions League football (2017/18 to 2022/23), and the COVID-19 pandemic, which almost removed matchday income in 2020/21.

Main research question

> To what extent are changes in Arsenal Football Club's financial resources and investment decisions associated with changes in sporting performance between 2015/16 and 2024/25?

Sub-questions (fixed before the analysis)

- **SQ1, financial evolution:** How did Arsenal's financial resources and financial structure change between 2015/16 and 2024/25?
- **SQ2, finance and investment:** How were changes in financial resources associated with changes in transfer investment?
- **SQ3, investment and later performance:** To what extent was transfer investment associated with later sporting performance?
- **SQ4, staff costs and later performance:** How were staff costs associated with later sporting performance?
- **SQ5, finance and later performance:** How were overall financial resources associated with changes in sporting performance?
- **SQ6, investment and team performance:** Was the scale of transfer investment associated with changes in underlying team performance?

I fixed these questions in the research design (`RESEARCH_DESIGN.md`) and added none after seeing the data. COVID-19 and the managerial eras are treated as context.

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2. DATA AND SOURCES

The panel (`arsenal_master_panel.csv`) has 10 rows, one for each season from 2015/16 to 2024/25, and 108 variables. I left out 2025/26 because it is not finished.

2.1 Financial data

- **Source:** audited annual reports of Arsenal Holdings (a plc until FY2018, a limited company after that) filed at Companies House.
- **Periods:** financial years ending 31 May, FY2016 to FY2025.

- **Main variables:** football turnover (matchday, broadcasting and commercial), staff costs, net debt, operating and net profit or loss, cash and player registration additions.
- **Staff costs are not player wages.** The accounts report one `staff_costs` figure for all employees: players, coaches, academy, medical, administrative and commercial staff, plus employer pension and social security charges. The accounts do not give a player-only wage bill.

2.2 Transfer data

- **Scope:** 128 transfer transactions (in and out) between June 2015 and May 2025.
- **Main measure:** permanent gross transfer spend, meaning the fixed fee agreed for permanent first-team signings, without loan fees or possible add-ons.
- **Measurement problem:** clubs usually do not disclose individual fees. Only 9 of the 128 transactions had a fee confirmed in a primary regulatory source. So 77.8% of total gross consideration (£845.50m of £1,087.03m over the decade) rests on secondary reporting (BBC Sport, The Athletic, Transfermarkt). The fee data therefore contain estimation error.

2.3 Sporting data

- **Official records:** Premier League tables and results (points, PPG, goal difference, league position).
- **Understat metrics:** expected goals (xG), expected goals against (xGA), expected goal difference (xGD), expected points (xPTS) and passes per defensive action (PPDA, where a lower number means more pressing). These are estimates from a third-party model, not official data.

2.4 Source tracking

Each cell in the panel is linked to a source in `master_provenance_map.csv`. Financial figures are checked against the audited accounts and match records against official records. Transfer fees and Understat metrics cannot be checked against official documents in the same way.

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3. METHOD

3.1 Why only correlations

With one club and N = 10, a regression with several variables would overfit, so I did not use multivariable regression or machine learning. I used bivariate correlations; for lagged tests I paired a financial variable in season t with a sporting outcome in season t+1.

3.2 Samples

1. **Same season (t with t):** N = 10.
2. **Lagged (t with t+1):** N = 9 pairs, from 2015/16 to 2016/17 up to 2023/24 to 2024/25.
3. **Without COVID:** N = 8 for same-season tests (dropping 2019/20 and 2020/21) and N = 6 for lagged tests (dropping the pairs 2018/19 to 2019/20, 2019/20 to 2020/21 and 2020/21 to 2021/22).

3.3 Measures

For each pair of variables I calculated Pearson's r (linear), Spearman's rank correlation and Kendall's tau-b. If all three had the same sign I marked the result `ALL_AGREE`; if the sign differed I marked it `SIGN_REVERSAL`. P-values are given only as a rough guide. With N = 9 or 10 and 17 tests, I did not correct for multiple testing, so a single p-value below 0.05 should not be read as a finding.

3.4 Sensitivity checks

1. **Leave-one-season-out (LOO):** I removed each season (or season pair) in turn and recalculated r, to see the range and whether the sign changes.
2. **COVID exclusion:** I recalculated r without the pandemic seasons.

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4. RESULTS

4.1 Descriptive statistics

Table 1: Descriptive statistics for the main variables, 2015/16 to 2024/25 (N = 10).

Variable	Mean	Median	Std Dev	Min (season)	Max (season)	Change, first to last season	% change
football_turnover (£m)	436.81	391.45	122.24	327.60 (20/21)	691.00 (24/25)	+340.40	+97.09%
staff_costs (£m)	245.07	233.10	51.37	195.40 (15/16)	347.00 (24/25)	+151.60	+77.58%
net_debt (£m)	134.58	154.00	120.45	-12.90 (17/18)	283.00 (24/25)	+276.50	N/A
operating_profit_loss (£m)	-36.55	-52.00	55.40	-130.40 (20/21)	+43.40 (16/17)	-81.50	N/A
net_profit_loss (£m)	-21.55	-27.65	47.03	-107.30 (20/21)	+56.50 (17/18)	-2.65	N/A
permanent_gross_transfer_spend (£m)	108.70	98.40	53.07	15.00 (15/16)	199.30 (23/24)	+73.85	+492.33%
permanent_net_transfer_spend (£m)	71.61	73.86	52.18	-8.50 (17/18)	141.70 (22/23)	+5.63	N/A
primary_transfer_spend_to_revenue_pct (%)	25.88	25.12	11.82	4.28 (15/16)	40.55 (19/20)	+10.65	+248.83%
points	71.20	70.50	10.06	56.00 (19/20)	89.00 (23/24)	+3.00	+4.23%
ppg	1.87	1.85	0.26	1.47 (19/20)	2.34 (23/24)	+0.08	+4.28%
league_position	4.50	5.00	2.42	2.00 (15/16)	8.00 (19/20)	+0.00	N/A
xgd	22.23	20.02	17.44	-6.43 (19/20)	52.61 (23/24)	-6.08	N/A
xpts	66.51	65.33	9.63	50.15 (19/20)	81.94 (23/24)	-4.05	-5.26%
ppda	10.04	9.51	1.81	8.26 (16/17)	13.64 (20/21)	+0.45	+5.39%

For league position the best value is the minimum (2nd place, reached in 2015/16 and again from 2022/23 to 2024/25) and the worst is 8th (2019/20 and 2020/21).

1. **Two financial periods.** The club made operating profits in the first three years (FY2016 to FY2018) and operating losses in every year after that. Turnover stayed between £327.60m and £422.80m until 2021/22, then rose to £691.00m by 2024/25, while net debt rose to £283.00m.
2. **Transfer spending.** Gross spending averaged £108.70m a season and peaked at £199.30m in 2023/24, the summer Arsenal signed Declan Rice and Kai Havertz.
3. **Low point and recovery.** Results were worst in 2019/20 (56 points, 1.47 PPG, 8th place, xGD of -6.43). They then recovered to 89 points, 2.34 PPG and an xGD of 52.61 in 2023/24.

4.2 Association results

Table 2: Correlations for the 17 pre-specified primary tests (SQ2 to SQ6). Lagged tests pair the predictor in season t with the outcome in season t+1.

Spec ID	Sub-question	Predictor (t)	Outcome	N	Pearson r	Spearman	Kendall tau	Agreement
PRIMARY-SQ2-01	SQ2	football_turnover	permanent_gross_transfer_spend (t)	10	+0.3659	+0.3455	+0.2444	ALL_AGREE
PRIMARY-SQ2-02	SQ2	staff_costs	permanent_gross_transfer_spend (t)	10	+0.3894	+0.3091	+0.2889	ALL_AGREE
PRIMARY-SQ2-03	SQ2	net_debt	permanent_gross_transfer_spend (t)	10	+0.5906	+0.4182	+0.3333	ALL_AGREE
PRIMARY-SQ3-04	SQ3	permanent_gross_transfer_spend	ppg (t+1)	9	+0.3911	+0.4500	+0.3333	ALL_AGREE
PRIMARY-SQ3-05	SQ3	permanent_gross_transfer_spend	xgd (t+1)	9	+0.5982	+0.6500	+0.5000	ALL_AGREE
PRIMARY-SQ3-06	SQ3	permanent_gross_transfer_spend	xpts (t+1)	9	+0.6189	+0.6667	+0.5000	ALL_AGREE
PRIMARY-SQ4-07	SQ4	staff_costs	ppg (t+1)	9	+0.0516	-0.0333	0.0000	SIGN_REVERSAL
PRIMARY-SQ4-08	SQ4	staff_costs	xgd (t+1)	9	+0.2229	+0.1833	+0.1667	ALL_AGREE
PRIMARY-SQ4-09	SQ4	staff_costs	xpts (t+1)	9	+0.2618	+0.2667	+0.1667	ALL_AGREE
PRIMARY-SQ5-10	SQ5	football_turnover	ppg (t+1)	9	+0.2412	+0.2333	+0.1667	ALL_AGREE
PRIMARY-SQ5-11	SQ5	football_turnover	xgd (t+1)	9	+0.5046	+0.5000	+0.3333	ALL_AGREE
PRIMARY-SQ5-12	SQ5	football_turnover	xpts (t+1)	9	+0.4851	+0.5167	+0.3333	ALL_AGREE
PRIMARY-SQ5-13	SQ5	net_debt	ppg (t+1)	9	+0.5663	+0.4333	+0.2778	ALL_AGREE
PRIMARY-SQ5-14	SQ5	net_debt	xgd (t+1)	9	+0.6320	+0.6333	+0.4444	ALL_AGREE
PRIMARY-SQ5-15	SQ5	net_debt	xpts (t+1)	9	+0.6937	+0.6167	+0.4444	ALL_AGREE
PRIMARY-SQ6-16	SQ6 (same as SQ3-05)	permanent_gross_transfer_spend	xgd (t+1)	9	+0.5982	+0.6500	+0.5000	ALL_AGREE
PRIMARY-SQ6-17	SQ6 (same as SQ3-06)	permanent_gross_transfer_spend	xpts (t+1)	9	+0.6189	+0.6667	+0.5000	ALL_AGREE

SQ6-16 and SQ6-17 are the same calculations as SQ3-05 and SQ3-06. I do not count them as extra evidence.

4.3 Sensitivity checks

Table 3: Leave-one-season-out (LOO) range and COVID-excluded Pearson r for the primary tests. SQ6-16 and SQ6-17 give the same results as SQ3-05 and SQ3-06 and are not repeated.

Spec ID	Base r	LOO min (season removed)	LOO max (season removed)	Sign change?	No-COVID r (N = 6 or 8)	Difference
PRIMARY-SQ2-01	+0.3659	0.0830 (2023/24)	0.6824 (2024/25)	No	0.4064	+0.0405
PRIMARY-SQ2-02	+0.3894	0.0751 (2023/24)	0.6768 (2024/25)	No	0.4179	+0.0286
PRIMARY-SQ2-03	+0.5906	0.4643 (2023/24)	0.7246 (2024/25)	No	0.6791	+0.0885
PRIMARY-SQ3-04	+0.3911	0.2411 (2022/23 to 2023/24)	0.6244 (2015/16 to 2016/17)	No	0.3818	-0.0093
PRIMARY-SQ3-05	+0.5982	0.5334 (2022/23 to 2023/24)	0.7157 (2015/16 to 2016/17)	No	0.6392	+0.0409
PRIMARY-SQ3-06	+0.6189	0.5560 (2022/23 to 2023/24)	0.7073 (2015/16 to 2016/17)	No	0.7061	+0.0872
PRIMARY-SQ4-07	+0.0516	-0.0936 (2023/24 to 2024/25)	0.1648 (2021/22 to 2022/23)	**Yes**	0.0924	+0.0408
PRIMARY-SQ4-08	+0.2229	-0.0909 (2023/24 to 2024/25)	0.3020 (2022/23 to 2023/24)	**Yes**	0.3348	+0.1120
PRIMARY-SQ4-09	+0.2618	-0.0298 (2023/24 to 2024/25)	0.3420 (2021/22 to 2022/23)	**Yes**	0.3810	+0.1192
PRIMARY-SQ5-10	+0.2412	0.1228 (2022/23 to 2023/24)	0.3591 (2021/22 to 2022/23)	No	0.0112	-0.2300
PRIMARY-SQ5-11	+0.5046	0.4692 (2019/20 to 2020/21)	0.5780 (2018/19 to 2019/20)	No	0.4607	-0.0439
PRIMARY-SQ5-12	+0.4851	0.4449 (2019/20 to 2020/21)	0.5651 (2018/19 to 2019/20)	No	0.4776	-0.0075
PRIMARY-SQ5-13	+0.5663	0.4841 (2021/22 to 2022/23)	0.6632 (2015/16 to 2016/17)	No	0.6839	+0.1176
PRIMARY-SQ5-14	+0.6320	0.5902 (2023/24 to 2024/25)	0.7214 (2016/17 to 2017/18)	No	0.7734	+0.1413
PRIMARY-SQ5-15	+0.6937	0.6593 (2023/24 to 2024/25)	0.7664 (2016/17 to 2017/18)	No	0.8335	+0.1398

5. DISCUSSION

5.1 SQ1: How the club's finances changed

What the data show. Football turnover rose from £350.60m in 2015/16 to £691.00m in 2024/25, an increase of £340.40m (97.09%). The rise was not steady. Turnover was between £388.20m and £422.80m from 2016/17 to 2018/19, fell to its lowest point of £327.60m in 2020/21, when matchday revenue dropped from £69.40m in 2019/20 to £3.80m, and then grew fast. The biggest single rise was in 2023/24: £151.40m (32.59%).

Staff costs rose by £151.60m (77.58%), from £195.40m to £347.00m. Net debt was £6.50m in 2015/16, became a net cash position of £12.90m (net debt of -£12.90m) in 2017/18, and then rose to £283.00m by 2024/25. Operating results averaged -£36.55m a season, with the largest loss, -£130.40m, in 2020/21.

My interpretation. I see three periods: self-financing at the end of the Wenger years, a difficult period without Champions League football and with the pandemic, and a rebuild paid for partly with the owner's money. The stadium bonds were replaced by a KSE loan and net debt rose a lot, so the club's greater spending power came together with more debt.

Other explanations. Part of the turnover growth may come from league-wide broadcast and commercial growth, not from Arsenal's own choices. The rise in net debt could be a response to losses outside the Champions League rather than a planned choice to borrow.

Limits. N = 10, and the COVID seasons pull down the averages for turnover and profit. Net debt includes the shareholder loan, so it reflects the owner's refinancing as well as normal borrowing. These trends do not show that one change caused another.

5.2 SQ2: Financial resources and transfer spending

What the data show. In the same season (N = 10), all three financial measures were positively linked to permanent gross transfer spend:

- Turnover: r = 0.3659 (p = 0.2984), Spearman = 0.3455, Kendall = 0.2444.
- Staff costs: r = 0.3894 (p = 0.2661), Spearman = 0.3091, Kendall = 0.2889.
- Net debt: r = 0.5906 (p = 0.0722), Spearman = 0.4182, Kendall = 0.3333.

All signs stayed positive in the LOO check. Removing 2023/24 gave the lowest r in all three tests (0.0830, 0.0751 and 0.4643) and removing 2024/25 the highest (0.6824, 0.6768 and 0.7246). Without the COVID seasons (N = 8), the correlations were a little higher (0.4064, 0.4179 and 0.6791).

My interpretation. The stronger link with net debt fits the idea that the big signing years were paid for partly with the owner's loans, not only from that year's income. The turnover link is moderate (r = 0.3659) and its LOO range is wide (0.0830 to 0.6824), so one or two seasons decide much of it.

Other explanations. Spending may have been driven by gaps in the squad and by changes of manager, more than by how much money the club had in a given year.

Limits. Fees are often paid in instalments over several years, but `permanent_gross_transfer_spend` records the whole fee in the season of the deal. Same-season correlations cannot tell whether revenue led to spending, whether spending led to borrowing, or whether a third factor (for example the 2018 ownership change) affected both.

5.3 SQ3: Transfer spending and next season's performance

What the data show. Across nine season pairs (three of them affected by COVID), spending in season t was positively linked to all three outcomes in season t+1:

- PPG: r = 0.3911 (p = 0.2980), Spearman = 0.4500, Kendall = 0.3333.
- xGD: r = 0.5982 (p = 0.0888), Spearman = 0.6500, Kendall = 0.5000.
- xPTS: r = 0.6189 (p = 0.0756), Spearman = 0.6667, Kendall = 0.5000.

In the sensitivity tests, net transfer spend was linked to next-season xGD (r = 0.7013) and xPTS (r = 0.6968), and gross spend had r = -0.5495 with next-season league position (a lower position number is better, so a negative sign means higher spending came before better finishes).

All three primary tests kept a positive sign in every LOO run (0.2411 to 0.6244 for PPG, 0.5334 to 0.7157 for xGD, 0.5560 to 0.7073 for xPTS). Removing the first pair, 2015/16 to 2016/17 (spending of £15.00m followed by 75 points), gave the highest r in all three. Without the COVID pairs (N = 6), r was 0.3818 for PPG, 0.6392 for xGD and 0.7061 for xPTS.

My interpretation. Spending is more closely linked to next season's xGD and xPTS (r of about 0.60 to 0.62) than to actual points (r = 0.3911). My reading is that new players may change the chances a team creates and allows more reliably than they change results, because results over 38 games also depend on luck and injuries. How well players are chosen and whether they fit the system probably matter too, but I did not measure these.

None of the Pearson p-values for these three tests is below 0.05 (the Spearman p-value for xPTS is 0.0499, just under it), and N is 9, so I treat this as a pattern worth testing on more clubs or more seasons, not as proof.

Other explanations. The improvement may come from coaching, squad stability or young players developing.

Limits. N = 9 (6 without COVID), and most fees are estimates (section 2.2). Spending coming one season earlier does not prove cause, and these correlations cannot control for coaching, analytics, medical staff or what rival clubs spent.

5.4 SQ4: Staff costs and next season's performance

What the data show. The links between staff costs in season t and performance in season t+1 (N = 9) were weak:

- PPG: r = 0.0516 (p = 0.8951), Spearman = -0.0333 (p = 0.9322), Kendall = 0.0000 (p = 1.0000).
- xGD: r = 0.2229 (p = 0.5643), Spearman = 0.1833, Kendall = 0.1667.
- xPTS: r = 0.2618 (p = 0.4962), Spearman = 0.2667, Kendall = 0.1667.

SQ4-07 is the only one of the 17 primary tests where the three methods disagree on the sign. In the LOO check, all three SQ4 tests became negative when the pair 2023/24 to 2024/25 was removed (staff costs of £327.90m in 2023/24, followed by 74 points and 1.95 PPG in 2024/25). The LOO ranges were -0.0936 to 0.1648 (PPG), -0.0909 to 0.3020 (xGD) and -0.0298 to 0.3420 (xPTS). Without COVID (N = 6), r was 0.0924, 0.3348 and 0.3810.

My interpretation. This is different from the multi-club studies, which find a strong link between wages and league position (for example Szymanski & Smith, 1997; Hall et al., 2002). I see two likely reasons:

1. **What the figure measures.** `staff_costs` covers every employee, not only players. More office, commercial or academy staff can raise it without changing the team on the pitch.
2. **Long contracts.** Player contracts last several years, so wages can stay high while results get worse. For example, staff costs rose to £244.40m in 2020/21, the highest level up to that point, while Arsenal finished 8th in both 2019/20 and 2020/21.

Other explanations. Studies of all 20 Premier League clubs compare title contenders with relegation sides, so wage gaps are very large; one top club's wage bill moves in a narrower range. Severance payments and wage inflation can also raise costs without improving the squad.

Limits. The weak positive result depends heavily on the last season pair. A near-zero correlation here does not mean player wages have no effect on results. It only means that total staff costs had no stable linear link with next-season points at this one club over these years.

5.5 SQ5: Financial resources and next season's performance

What the data show. With N = 9 lagged pairs:

- Turnover and PPG: r = 0.2412 (p = 0.5318), Spearman = 0.2333, Kendall = 0.1667.
- Turnover and xGD: r = 0.5046 (p = 0.1659), Spearman = 0.5000, Kendall = 0.3333.
- Turnover and xPTS: r = 0.4851 (p = 0.1856), Spearman = 0.5167, Kendall = 0.3333.
- Net debt and PPG: r = 0.5663 (p = 0.1119), Spearman = 0.4333, Kendall = 0.2778.
- Net debt and xGD: r = 0.6320 (p = 0.0678), Spearman = 0.6333, Kendall = 0.4444.
- Net debt and xPTS: r = 0.6937 (p = 0.0382), Spearman = 0.6167, Kendall = 0.4444.

All six kept a positive sign in the LOO check. Without COVID (N = 6), the turnover and PPG correlation fell from 0.2412 to 0.0112 (a change of -0.2300), while the net debt correlations rose (0.6839 for PPG, 0.7734 for xGD, 0.8335 for xPTS).

My interpretation. More money did not simply mean better results. The turnover and PPG link almost disappears without the COVID seasons, so I think most of it came from revenue and results both being low in 2019/20 and 2020/21.

The net debt results look strong, but I am careful with them. Net debt rose over most of the decade, and results got better in the second half, so two series that both trend upward over the same years will be correlated even if one does not affect the other. My reading is that the club took on debt to

rebuild the squad before the recovery of 2022/23 to 2023/24, but these data cannot test that. The causation could also run the other way: better results bring more broadcast, prize and commercial income.

Net debt and xPTS (p = 0.0382) is the only primary test with a Pearson p-value below 0.05. With this many tests and N = 9, one such result is not strong evidence.

Limits. N = 9 (6 without COVID), and the 2018 ownership change and refinancing are specific to Arsenal, so these results may not apply to other clubs. Debt funds spending; it does not produce points.

5.6 SQ6: Transfer spending and underlying team performance

What the data show. SQ6 uses the same calculations as SQ3 for xGD (r = 0.5982, p = 0.0888) and xPTS (r = 0.6189, p = 0.0756). I keep it in the paper because it was in the pre-specified design, but it is not separate evidence. For context, PPDA fell from 13.64 in 2020/21 to 8.80 in 2024/25 (more pressing), and xGD rose from -6.43 in 2019/20 to 52.61 in 2023/24.

My interpretation. The improvement in xGD and pressing came in the same years as large transfer spending, which is consistent with new signings helping the change in playing style. But coaching, training and team cohesion changed in the same years too.

Other explanations. The better numbers may come from the same manager staying for several seasons and the team learning his system, with new players joining a structure that already worked.

Limits. The panel has no data on minutes played by new signings or squad age, so I cannot separate what new players contributed from what existing players contributed. The results show that spending and team-level metrics moved together, not that spending caused the improvement.

5.7 Putting the results together

My reading of the results is that money makes squad investment possible but does not decide results on its own. Figure 1 shows the framework I use to think about this. It is a hypothesis, not something the correlations test.

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Financial resources (SQ1): turnover, cash, borrowing
        |
        v
Investment decisions (SQ2): transfer spending, how it is financed
        |
        v
Squad inputs (SQ4, SQ6): staff costs, quality and fit of signings
        |
        v
Sporting performance (SQ3, SQ5): points, xGD, xPTS
        |
        +--> back to financial resources (prize money, matchday demand, commercial income)
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Figure 1: Conceptual framework: money enables investment, and results feed back into income. Not tested in this paper.

1. **How money is spent may matter more than how much.** Total staff costs had a weak link with later results (SQ4), while transfer spending had a stronger link with later xGD and xPTS (SQ3). I read this as a sign that allocation matters, but these data cannot measure allocation quality directly.
2. **Feedback.** Success probably brings more income (Champions League money, commercial deals), so finance and results may affect each other. This would also make simple correlations hard to interpret. I did not test this direction.

5.8 Limitations

1. **Small sample, one club.** N = 10 same-season, N = 9 lagged and N = 6 lagged without COVID. This is too few for regression or causal methods, and with 17 tests some correlations will look large by chance.
2. **Estimated transfer fees.** Only 9 of 128 transactions had a fee confirmed in a primary source, and 77.8% of gross consideration (£845.50m of £1,087.03m) comes from secondary reports.
3. **Staff costs are not player wages.** The accounts give one figure for all staff, so I cannot separate first-team wages from other pay.
4. **Understat data.** xG, xPTS and PPDA come from a third-party model, not official records.
5. **COVID.** In 2019/20 and 2020/21, empty stadiums and a compressed fixture list affected both the financial and the sporting figures.

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6. CONCLUSION

Using a ten-season panel built from audited accounts, a census of 128 transfers and match data, I found:

1. **Transfer spending and next season's expected performance.** Spending was positively linked to next-season xGD (r = 0.5982) and xPTS (r = 0.6189), and less to next-season PPG (r = 0.3911). The signs held in all LOO runs, but N = 9 and no Pearson p-value was below 0.05.
2. **No stable link for staff costs.** Staff costs had almost no link with next-season PPG (r = 0.0516, Spearman = -0.0333), and the sign changed in the LOO check. I think this partly reflects that the figure covers all staff, not only players, and that contracts last several years.
3. **Spending and borrowing.** Same-season transfer spending was more closely linked to net debt (r = 0.5906) than to turnover (r = 0.3659). I read this as the club relying on the owner's loan during the rebuild, but both series also share a time trend.
4. **Possible feedback.** Results probably also affect income, so the link between money and results may run in both directions. I did not test this.

All of these are correlations in one club over ten seasons. They describe what moved together; they do not show that spending caused better results.

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7. DATA FILES

All data and files are public at github.com/ege211/sports-finance-performance:

- `data/arsenal_master_panel.csv`: the panel (10 rows by 108 variables).
- `data/financial_provenance.csv` and `data/master_provenance_map.csv`: where each figure comes from (Arsenal Holdings annual reports, Companies House, 2016 to 2025; transfer census; Premier League records and Understat).
- `results/`: the descriptive, association and sensitivity results.

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